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SCIENTIFIC ARTICLE

Percutaneous cryoablation of musculoskeletal tumours: the largest reported single-centre case series — outcomes, complications and subgroup analysis

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Abstract

Objectives Cryoablation is an image-guided, minimally invasive treatment technique that uses extreme cold to induce tumour destruction. Its role in musculoskeletal oncology has expanded in recent years, although large clinical series remain limited. This study reports a large single-centre experience of percutaneous cryoablation for musculoskeletal tumours, including metastatic lesions, benign and cartilaginous tumours, and primary bone tumours.

Materials and methods This was a single-centre retrospective cohort study conducted at a tertiary musculoskeletal oncology centre. All procedures were performed by a single specialist musculoskeletal interventional radiologist with over 20 years of experience. The study included patients who underwent percutaneous cryoablation for musculoskeletal tumours between June 2021 and December 2025. Clinical response, radiological response, recurrence, and complications were assessed over a median follow-up of 3 months (mean 8.7 months; range 0–48 months). Radiological response was assessed using pre-treatment and post-treatment Hounsfield Unit (HU) measurements on CT; this is not a validated radiological scoring system and is reported as a complementary, exploratory marker alongside standard clinical response assessment. Complications were graded using the CIRSE classification system.

Results A total of 48 patients underwent percutaneous cryoablation during the study period. The cohort included 26 males and 22 females, with a median age of 40 years. A good or complete response was achieved in 28 patients, representing 58% of the total cohort. A partial response was achieved in seven patients, representing 15% of the total cohort. HU reduction was observed in 45 of 48 patients, representing 94% of the cohort. The overall complication rate was 6.3%, with three complications recorded among 48 procedures. No major complications or procedure-related deaths were observed.

Conclusions Cryoablation appears to be a safe and effective minimally invasive treatment for selected musculoskeletal tumours. Most patients achieved a good, complete, or partial response, with a low complication rate. Hounsfield Unit reduction after treatment may provide a useful, though as yet non-validated, marker of radiological response. Further prospective studies with longer follow-up, standardised response criteria and validated complication grading are needed to confirm these findings.

Keywords Oncology · Artificial intelligence · Musculoskeletal · Cryotherapy · Interventional radiology

Background

Role of image-guided ablation in musculoskeletal oncology

Image-guided percutaneous ablation has become an increasingly important treatment option in musculoskeletal oncology [1]. It offers a minimally invasive alternative to open surgery in selected patients. This is particularly valuable for patients who are poor surgical candidates, have anatomically

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challenging lesions, or require symptom control rather than curative resection [2].

Ablative techniques can be used for benign tumours, primary bone tumours, and skeletal metastases. However, the main aims of treatment vary according to pathology. In benign lesions, the aim is often local disease control and symptom relief [3]. In metastatic disease, the aim is frequently palliation, pain reduction, and preservation of function [4].

Principles and advantages of cryoablation

Cryoablation is a thermal ablation technique that destroys tumour tissue through controlled freezing [5]. During the procedure, cryoprobes are inserted into the target lesion under image guidance. The probes generate an ice ball that progressively expands throughout the lesion [6, 7] (Fig. 1). Cell death occurs through several mechanisms, including intracellular ice crystal formation, osmotic injury, membrane destruction, and vascular thrombosis [8, 9]. Repeated freeze–thaw cycles increase the extent of tissue destruction [9].

A key advantage of cryoablation is that the ice ball can be visualised during treatment. This allows the operator to monitor the ablation zone in real time. Real-time visualisation may improve procedural accuracy and reduce injury to adjacent structures [10]. Cryoablation also has an intrinsic analgesic effect. This may make it particularly useful for patients with painful skeletal lesions and metastatic bone disease [11].

Relevance to musculoskeletal tumours

Musculoskeletal lesions are frequently located close to joints, nerves, vessels, skin, or weight-bearing structures, so precise control of the ablation zone is paramount. Cryoablation is particularly well suited to this setting for two related reasons: the ice ball is directly visible as a low-attenuation zone on CT, allowing the treatment margin to be defined and monitored in real time with a precision not offered by heat-based modalities, whose ablation zone is not directly visualised during treatment [12, 13]; and protective manoeuvres such as hydrodissection and thermal insulation can be used alongside this real-time monitoring to actively displace or shield adjacent critical structures [12].

In selected cases, this direct visibility of the treatment margin may make cryoablation preferable to other thermal techniques on CT [13], although this study did not perform a direct comparison between modalities and no superiority claim is made here. Cryoablation may also be useful when surgery would carry a high risk of morbidity. In benign and cartilaginous lesions, it may reduce the need for more invasive curettage or resection [14]. In metastatic disease, it may provide local tumour control and rapid pain relief [11, 15]. It may also be used alongside systemic therapy, radiotherapy, or surgical stabilisation when required [16].

Current evidence base

The published evidence supporting musculoskeletal cryoablation has increased in recent years [1, 17]. However, much of the existing literature remains limited by small



Fig. 1 Cryoablation needles in situ, ice ball formation

cohort sizes [18]. Many studies focus on specific diagnoses, such as osteoid osteoma, chondroblastoma, or painful skeletal metastases [19–21]. Fewer studies have evaluated outcomes across a broad range of musculoskeletal tumour types. There is also limited evidence comparing outcomes between tumour subgroups. As a result, the role of cryoablation across heterogeneous musculoskeletal oncology practice remains incompletely defined [22].

Need for objective radiological assessment

Clinical improvement is an important outcome after cryoablation. However, a narrative clinical assessment of improvement alone may not fully reflect the local treatment effect and is inherently subject to observer variability, particularly in a retrospective series where no standardised pain or functional score was applied prospectively. Objective imaging markers are therefore useful when assessing treatment response.

Hounsfield Unit (HU) analysis on CT may provide a quantitative method for evaluating radiological change after ablation [23, 24]. A reduction in lesion density after treatment may reflect necrosis, cystic change, or fatty replacement. HU change may therefore provide a measurable marker of ablation response [25]. Unlike Response Evaluation Criteria in Solid Tumours (RECIST) or modified RECIST, which are validated for measurable soft-tissue tumour burden assessed by unidimensional diameter change, no validated HU-based scoring system or response threshold currently exists for osseous lesions treated with ablation; this approach is therefore presented as an exploratory, complementary marker rather than a substitute for standardised response criteria, and may be particularly useful in retrospective studies where standardised clinical outcome scores are not always available.

Study rationale and aim

There remains a need for larger clinical series evaluating musculoskeletal cryoablation in real-world oncology practice. There is also a need to assess outcomes across different tumour subgroups [26]. This study reports a single-centre case series of patients who underwent percutaneous cryoablation for musculoskeletal tumours. The study includes metastatic lesions, benign and cartilaginous tumours, and primary bone tumours.

The aim was to evaluate clinical response, radiological response, recurrence, and complications following treatment. A further aim was to explore whether treatment response differed between tumour subgroups. Radiological response was assessed using pre-treatment and post-treatment HU measurements.

Methods

Study design and patient selection

This was a single-centre retrospective cohort study conducted at a tertiary musculoskeletal oncology centre. Patients were identified from a departmental procedural database. The study included patients who underwent percutaneous cryoablation for musculoskeletal tumours between June 2021 and December 2025. Eligible patients had complete demographic data, histological confirmation of diagnosis, and at least one post-procedural imaging assessment, at a median follow-up of 3 months (mean 8.7 months; range 0–48 months). Histology was obtained either before cryoablation or during the same procedural session when required. All cases were reviewed through the local multidisciplinary team before treatment, and written informed consent was obtained before each procedure.

Cryoablation technique

Cryoablation was performed by a single specialist musculoskeletal interventional radiologist with over 20 years of experience in orthopaedic oncology intervention. Procedures were performed under CT guidance, ultrasound guidance, or a combination of both, depending on lesion location and visibility. Ultrasound was used for superficial or soft-tissue accessible locations as well as initial percutaneous access. CT was used for probe placement and ice ball monitoring in deeper and osseous lesions. Cryoprobes were inserted into the target lesion using aseptic technique. The number and arrangement of probes were selected according to lesion size, morphology, and anatomical location. All planning was performed on paper, by the radiologist, using the manufacturer's ice ball diameters and measurements.

A standard freeze–thaw protocol was used, typically consisting of two 10-min freeze cycles followed by passive and active thawing [9]. The ice ball was monitored during treatment to confirm adequate lesion coverage and reduce injury to adjacent structures. Protective techniques, including hydrodissection and warm saline infusion, were used when lesions were close to skin, nerves, vessels, joints, or other critical structures.

Subgroup classification

The cases were split into three groups: (1) Skeletal Metastases (14 cases), which were confirmed secondary bone diseases from any primary source, including renal cell carcinoma (5 cases), breast, lung, HPB, oropharyngeal, salivary gland, and angiosarcoma; (2) Benign and

Cartilaginous Lesions (23 cases), which included chondroblastoma (and any related metastasis treated with curative intent), chondromyxoid fibroma (CMF), chondroid lesions, cartilage tumours, atypical cartilaginous tumour (ACT), fibrous dysplasia, fibrous histiocytoma, haemangioma, and simple bone cysts; and (3) Others (10 cases), including osteoblastoma, aneurysmal bone cysts (ABC), giant cell tumours (GCT), arteriovenous malformations (AVM), non-oncological lesions, and one case that couldn't be categorised. These counts were double-checked against the operator's original records to ensure accuracy (Mets = 14, Benign/Cartilaginous = 23, Others = 10).

Outcome measures

The primary outcome measured was clinical response after cryoablation, assessed using clinical documentation and follow-up imaging. Treatment response was classified as good or complete, partial, or poor. A good or complete response was defined as marked symptomatic improvement together with supportive imaging findings (radiological necrosis, fatty change, consolidation, or cystic change), or clinical resolution. A partial response was defined as incomplete treatment effect, with residual symptoms or residual viable tumour. This composite clinical response assessment, rather than the HU-based radiological response described below, is what incorporates the full range of qualitative radiological signs (necrosis, fatty change, consolidation, cystic change) referred to elsewhere in this manuscript.

Radiological response was assessed separately, as a single objective and reproducible quantitative marker applied uniformly across the whole cohort regardless of underlying pathology, using HU measurements on CT. Pre-treatment HU was measured on the last CT before treatment, and post-treatment HU immediately following the procedure. A single region of interest (ROI) was placed freehand over the lesion on the same axial level pre and post treatment. All measurements were taken by a single reader on the institution's PACS system, hence no agreement statistics were performed. A reduction in HU was interpreted as reduced lesion density after treatment; there was no pre-determined criteria for this and was treated as exploratory. Additionally, this HU-based metric was not stratified by histological subtype for this analysis. The treated lesions characteristically contained a soft-tissue component that was observed to decrease in size on follow-up imaging after cryoablation, consistent with the reduction in HU. All soft tissue cases reduced in lesion size ton 0 mm. Secondary outcomes included recurrence, complication rate, and follow-up status. Complications were recorded from clinical records and graded according to the CIRSE classification system [27].

Statistical analysis

Statistical analysis was performed using Python. Continuous variables were summarised using means and standard deviations, or medians and ranges where appropriate. Categorical variables were reported as frequencies and percentages.

Paired pre-treatment and post-treatment HU values were compared using the Wilcoxon signed-rank test. Between-group comparisons were performed using non-parametric tests, and categorical associations were assessed using chi-squared testing. A two-tailed p-value of less than 0.05 was considered statistically significant.

Ethics and AI statement

This study was registered as a retrospective audit. As per the institution's guidelines, this did no require formal ethical approval. The authors declare the use of artificial intelligence for improving the wording of this manuscript with all outputs verified.

Results

Patient demographics

Forty-eight patients had cryoablation during the study period (Fig. 2). The group included 26 men (54.2%) and 22 women (45.8%). The average age was 42.5 years, with a median of 40 and a range of 16 to 89 years (Table 1).

There was a clear difference in age between the subgroups ($p = 0.0003$). Patients with metastases were the oldest, with an average age of 61.6 years (median 62). In contrast, those in the Benign and Cartilaginous Lesions and 'Others' groups were significantly younger, with averages of 34.1 and 35.4 years, respectively (medians of 28 and 27).

The femur was the most common site for treatment (11 out of 48, or 23%), followed by the humerus (8%) and the ilium (8%). The number of cases grew from 2021 to 2025.

Treatment response

Clinical response data were available for 39 out of 48 patients (Table 2). Across all 48 cases, 28 (58%) had a good or complete response, 7 (15%) had a partial response, and one (2%) had a poor response. Three patients (6%) died from their cancer spreading before the follow-up could take place (Fig. 3).

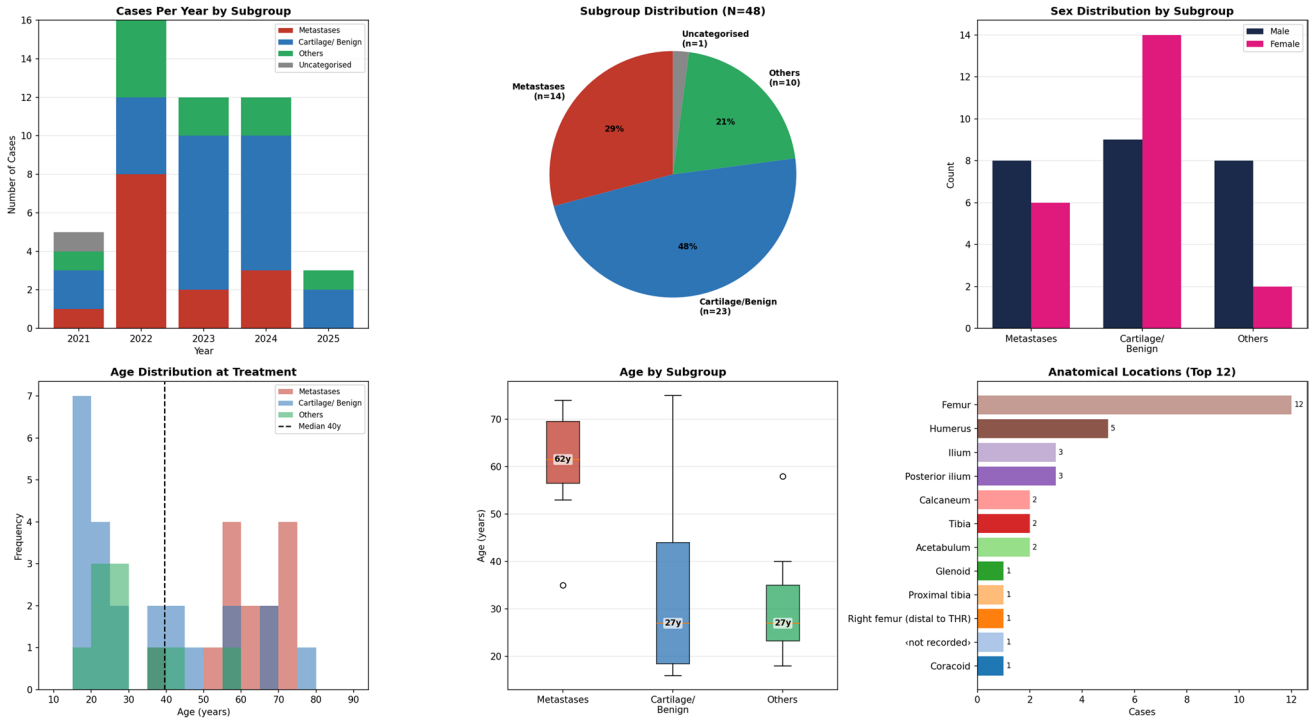


Fig. 2 Patient demographics and case distribution

Table 1 Patient demographics and lesion characteristics

Characteristic	Overall ($N=48$*)	Metastases ($n=14$)	Benign/Cartilaginous ($n=23$)	Others ($n=10$)
Age, mean ± SD	42.5 ± 20.8	61.6 ± 10.4	34.2 ± 18.0	30.5 ± 11.2
Age, median (range)	40 (16–89)	62 (41–89)	27 (16–75)	27 (18–58)
Male, n (%)	26 (54%)	8 (57%)	9 (39%)	8 (80%)
Female, n (%)	22 (46%)	6 (43%)	14 (61%)	2 (20%)
Most common site	Femur (23%)	Femur/Ilium	Femur	Femur
Date range	2021–2025	2021–2025	2021–2025	2021–2025
Median follow-up, months (range)	3 (0–48)	1 (0–36)	4 (1–48)	4 (1–36)

*1 case had indeterminate histology. It is included in the overall cohort for completeness but is absent in the subgroup analysis

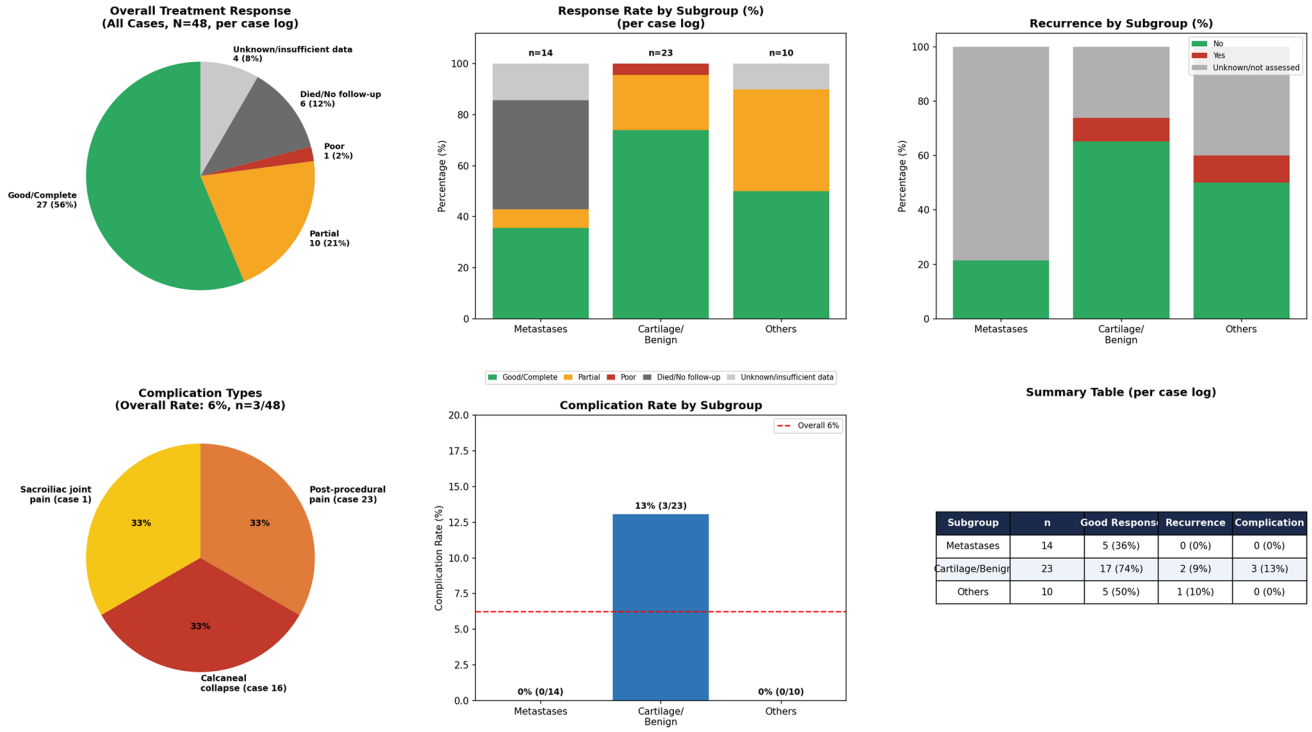
Table 2 Clinical outcomes by tumour subgroup

Subgroup	n	Good/Complete	Partial	Poor	No F/U or Died	Recurrence (known)	Complications
Metastases	14	5 (36%)	1 (7%)	0 (0%)	3 (21%)	0/3 known	0 (0%)
Benign/Cartilaginous	23	17 (74%)	5 (22%)	1 (4%)	0 (0%)	2/13 known (cartilage tumour, calcaneum; chondroblastoma, humerus)	3 (13%)
Others	10	5 (50%)	4 (40%)	0 (0%)	1 (10%)	1/9 known (osteoblastoma)	0 (0%)
Overall	48*	28 (58%)	7 (15%)	1 (2%)	3 (6%)	3/24 known	3 (6.3%)

*1 case had indeterminate histology. It is included in the overall cohort for completeness but is absent in the subgroup analysis

The Benign and Cartilaginous Lesions group had the best results, with a 78% good response rate (14 out of 18). This was followed by the 'Others' group at 56% (9 out of 16) and the Metastases group at 36% (5 out of 14).

The difference in response rate between subgroups approached, but did not reach, statistical significance ($p=0.056$).



Subgroup	n	Good Respons	Recurrence	Complication
Metastases	14	5 (36%)	0 (0%)	0 (0%)
Cartilage/Benign	23	17 (74%)	2 (9%)	3 (13%)
Others	10	5 (50%)	1 (10%)	0 (0%)

Fig. 3 Treatment outcomes—Response, Recurrence, and Complications

Subgroup analysis

Skeletal metastases (14 cases)

This group had a median age of 62 years (range 41–89). The cancers had come from various places, including renal clear cell carcinoma (5 cases, 36%), unknown primaries (2 cases, 14%), and single cases of hepato-pancreato-biliary (HPB) carcinoma, oropharyngeal, lung, breast, salivary duct carcinoma, angiosarcoma, and an EBV-related smooth muscle tumour (Fig. 4 and Table 3). All procedures were done to ease symptoms (palliative intent). Three patients died within 12 to 18 months due to their systemic disease getting worse, but none of these deaths were caused by the procedure. No complications occurred in this group. Every single patient showed a drop in HU on their scans.

Benign and cartilaginous lesions (23 cases)

This group had the best results, with a 78% good response rate and no complications. This was the youngest group (median age 27 years) and included chondroblastoma (6 cases, including one that recurred—see below—plus a separate case of metastatic chondroblastoma treated with curative intent), chondromyxoid fibroma (CMF, 2 cases), cartilage tumours (5 cases), chondroid lesions (2 cases), atypical cartilaginous tumour (ACT, 1 case), fibrous dysplasia (2 cases), fibrous histiocytoma (1 case), simple bone cysts (1 case), and haemangiomas (2 cases, one epithelioid).

The femur, ilium, and humerus were the most common sites. One chondroblastoma (talus) came back approximately 21 months after initial presentation. This group contained all 3 complications.

Others—including primary bone tumours (10 cases)

This was the most varied group, featuring osteoblastoma (3 cases, including one recurrence treated with repeat cryoablation and one mixed osteoblastoma/osteosarcoma that needed surgery later). It also included aneurysmal bone cyst (ABC, 2 cases), giant cell tumour (GCT, 2 cases), arteriovenous malformation (AVM, 1 case), a non-oncological lesion (1 case), and a giant cell reparative granuloma (1 case).

Recurrence

Recurrence data were available for 24 patients. Three patients developed local or systemic recurrence during follow-up. This represented 12.5% of patients with available recurrence data. The three recorded recurrences were:

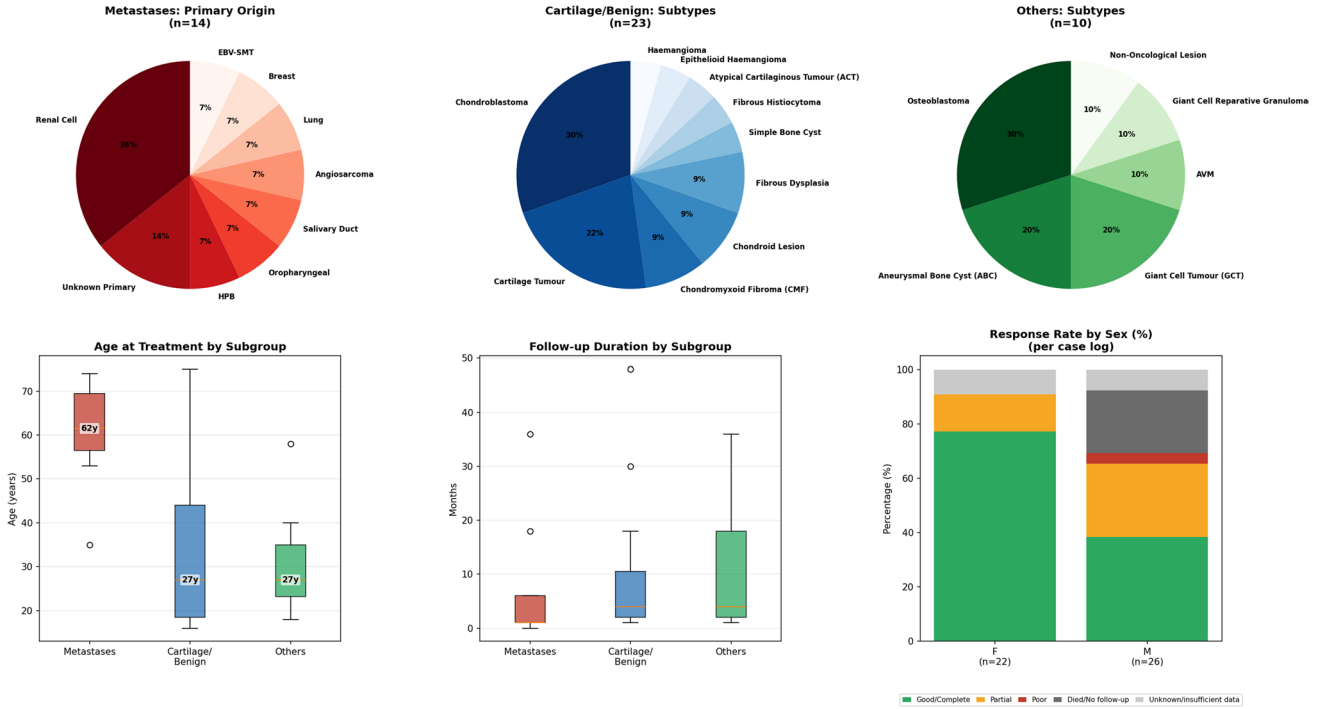


Fig. 4 Subgroup analysis—diagnostic distribution and outcomes

a cartilage tumour of the calcaneum (the same case as the calcaneal-collapse complication with no repeat cryoablation recorded), an osteoblastoma (retreated with repeat cryoablation), and a chondroblastoma with a poor response (recurrence recorded, no repeat treatment recorded).

Complications

The overall complication rate was 6.3% (3 of 48 procedures). All three complications were graded using the CIRSE classification system [27] (Table 4). No major complications were reported, and no procedure-related mortality occurred.

One patient developed mild sacroiliac joint pain after posterior iliac cryoablation with no therapy required (CIRSE grade 1). Another patient developed calcaneal collapse after treatment of a cartilage tumour, managed without surgery, received bracing and immobilisation (CIRSE grade 3). A further patient developed self-limiting post-procedural pain at the ablation site after treatment of chondroblastoma metastasis, which is consistent with CIRSE grade 1.

Discussion

Principal findings

This study reports a large single-centre experience of percutaneous cryoablation for musculoskeletal tumours. The findings suggest that cryoablation is a safe and effective treatment option across a mixed musculoskeletal oncology cohort. A good or complete clinical response was achieved in over half of patients. A further proportion of patients achieved a partial response after treatment. The overall complication rate was low, and no major complications were recorded. These findings support the use of cryoablation as a minimally invasive option for selected musculoskeletal tumours.

Clinical response

Clinical response was favourable in most patients with available follow-up data. The strongest clinical response was observed in the Benign and Cartilaginous Lesions subgroup, which was also the youngest group in the cohort—consistent with the typical age patterns for these conditions (metastatic disease predominantly affecting older patients; chondroblastoma and many benign cartilaginous or fibro-osseous lesions occurring predominantly in children and young adults). This may reflect the contained nature of many benign bone lesions and their suitability for local ablative treatment.

Subgroup interpretation

The Benign and Cartilaginous Lesions subgroup had the highest rate of good or complete response. This suggests

Table 3 Diagnosis (entity-level) distribution across the cohort, by subgroup

Subgroup	Diagnosis	n
Skeletal Metastases ($n=14$)		
	Renal cell carcinoma; clear cell carcinoma	5
	Unknown primary	2
	Hepato-pancreato-biliary (HPB) carcinoma	1
	Oropharyngeal carcinoma	1
	Lung carcinoma	1
	Breast carcinoma	1
	Salivary duct carcinoma	1
	Angiosarcoma	1
	EBV-related smooth muscle tumour	1
Benign/Cartilaginous ($n=23$)		
	Chondroblastoma, primary (including 1 that recurred, not re-treated with cryoablation)	6
	Chondroblastoma, metastatic, treated with curative intent	1
	Chondromyxoid fibroma (CMF)	2
	Cartilage tumours (incl. 1 with calcaneal collapse complication)	5
	Chondroid lesions	2
	Atypical cartilaginous tumour (ACT)	1
	Fibrous dysplasia	2
	Fibrous histiocytoma	1
	Simple bone cyst	1
	Haemangioma (incl. 1 epithelioid)	2
Others (n = 10)		
	Osteoblastoma (incl. 1 recurrence retreated with repeat cryoablation	3
	Aneurysmal bone cyst (ABC)	2
	Giant cell tumour (GCT)	2
	Arteriovenous malformation (AVM)	1
	Non-oncological lesion	1
	Giant cell reparative granuloma	1
Uncategorised ($n=1$)		
	Histologically indeterminate	1

that cryoablation may be particularly useful for selected benign and cartilaginous lesions. In these patients, cryoablation may reduce the need for more invasive surgery, such as curettage or resection. This may be especially important in younger patients or in lesions close to joints and growth-related structures.

In the metastatic subgroup, treatment was mainly palliative. The aims of treatment in metastatic disease were pain relief, local tumour control, and preservation of function. The lower response rate in this subgroup approached statistical significance ($p=0.056$; Sect. "Treatment Response"); metastatic deposits often have infiltrative, poorly defined margins that are harder to encompass completely within an ablation zone, compared with the well-defined margins typical of benign cartilaginous lesions, which may explain this difference. Although clinical response rates were lower in the metastatic subgroup, interpretation is limited by systemic disease progression and shorter follow-up. Several patients in this group died from progressive malignancy, rather than treatment-related causes. All patients in this subgroup showed a reduction in HU on follow-up imaging, which may indicate a local ablative effect even where systemic disease continued to progress.

The primary bone tumour subgroup was small, and conclusions from this subgroup should therefore be interpreted cautiously.

Safety and complications

Cryoablation was associated with a low complication rate in this cohort. All three complications were graded using the CIRSE classification system [27] (Table 3); with 2 CIRSE grade 1 and 1 CIRSE grade 3. No major complications,

Table 4 Complications and recurrence after cryoablation, graded using the CIRSE classification system, and recurrence after cryoablation

#	Subgroup	Complication	Management described	Proposed CIRSE grade
1	Benign/Cartilaginous	Mild sacroiliac joint pain after treatment of chondromyxoid fibroma (posterior ilium)	No analgesia needed	Grade 1
2	Benign/Cartilaginous	Calcaneal collapse after treatment of a cartilage tumour	Bracing needed but no surgery	Grade 3
3	Benign/Cartilaginous	Self-limiting post-procedural pain at ablation site after treatment of chondroblastoma metastasis	Resolved spontaneously, no therapy	Grade 1
Outcome		Finding		
Overall complication rate		6.3% (3/48)		
Number of complications		3/48		
CIRSE grade range		1–3		
Major complications		0		
Procedure-related mortality		0		
Recurrence data available		24 patients		
Recurrence rate among evaluable patients		3/24, 12.5%		
Repeat cryoablation required		1 case (osteoblastoma)		

neurovascular injuries, infections, or procedure-related deaths were observed. This supports the safety of percutaneous cryoablation when performed by an experienced specialist musculoskeletal interventional radiologist.

The ability to visualise the ice ball during treatment may contribute to procedural safety. Real-time monitoring allows the operator to assess the ablation margin and avoid critical adjacent structures. Protective techniques, such as hydrodissection and thermal protection, may further reduce procedural risk (Figs. 5 and 6).

Comparison with existing practice

Cryoablation offers several advantages compared with more invasive surgical approaches. It can be performed percutaneously and may reduce recovery time, morbidity, and hospital stay. It may also be valuable for patients who are unsuitable for surgery due to comorbidity, disease burden, or lesion location [1].

Cryoablation may allow more direct visualisation of the treatment zone than some other thermal ablation techniques, based on differences in imaging characteristics reported elsewhere [13], which may be advantageous in musculoskeletal lesions located near joints, nerves, vessels, or skin. This study did not perform a direct comparison with other ablative modalities, so this observation should not be read as a demonstrated superiority claim. Cryoablation can also be integrated with other treatments, including systemic therapy, radiotherapy, and surgical stabilisation [12].

Limitations

The study has some important limitations. The retrospective design introduces the risk of selection bias and incomplete data capture. The cohort was heterogeneous, including benign, primary malignant, and metastatic lesions grouped, in the case of the Benign and Cartilaginous Lesions subgroup, on the basis of shared clinical behaviour rather than shared histogenesis. A standardised, validated objective measure of clinical improvement (e.g. a numerical pain rating or functional outcome score) was not applied; response was instead determined from narrative clinical documentation and imaging review, which is inherently subject to observer variability. The HU-based radiological response metric used here is not a validated, externally benchmarked scoring system (unlike RECIST/mRECIST for measurable soft-tissue disease), and the precise threshold denoting a meaningful response requires prospective validation; response was also not stratified by individual histological diagnosis, only by the three broader subgroups. Recurrence data were only available for a subset of patients, which limits interpretation of long-term disease control. The small size of some subgroups, particularly the primary bone tumour subgroup, limits the strength of subgroup comparisons.

Future directions

Future prospective studies are required to validate these findings. Larger multicentre cohorts would improve the

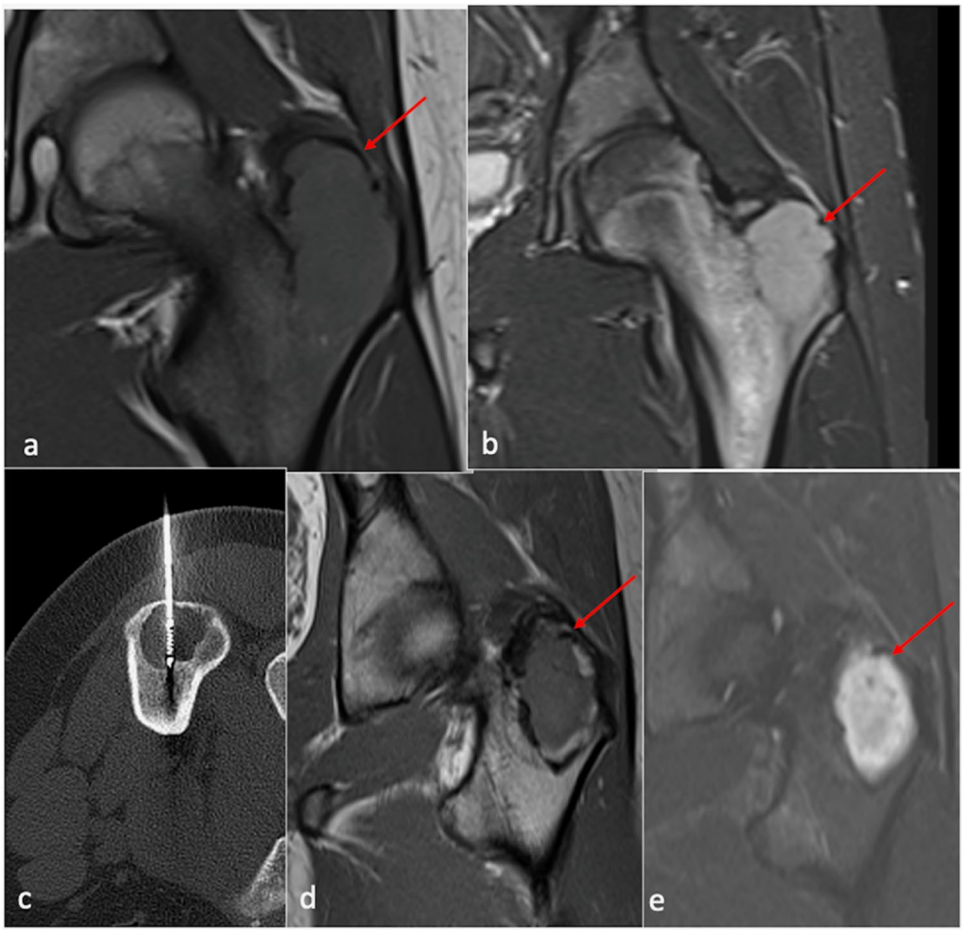


Fig. 5 Axial T1 (**a**), STIR (**b**) shows chondromyxoid fibroma (arrow), CT guided cryotherapy (**c**) and axial T1 (**d**) and STIR (**e**) shows marked consolidation with fatty change and cystic change

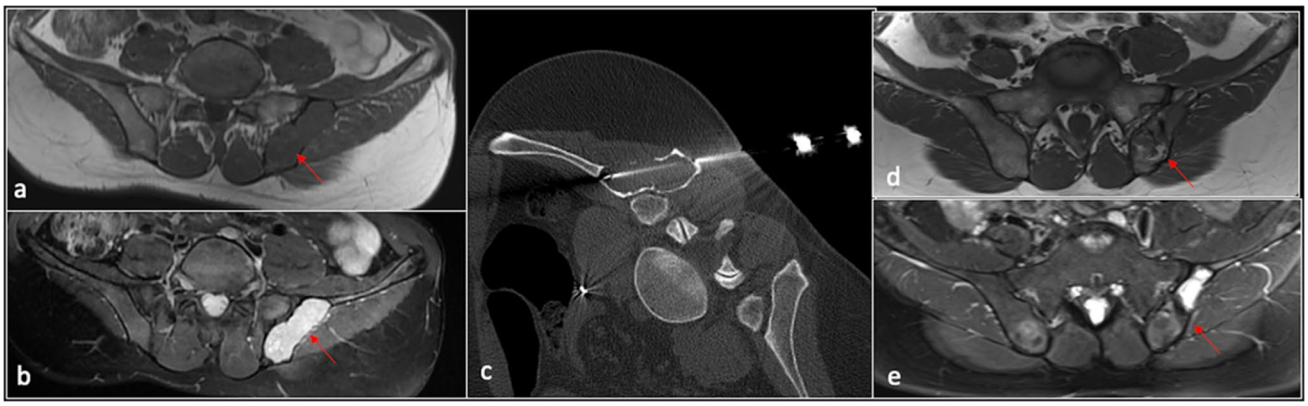


Fig. 6 Coronal T1 (**a**), T2 fat suppressed (**b**) shows chondroblastoma (arrow), CT guided cryotherapy (**c**) and coronal T1 (**d**) and STIR (**e**) shows marked consolidation of chondroblastoma with fatty change and cystic change

assessment of tumour-specific outcomes. Standardised pain scores, functional outcome measures, and quality-of-life assessments should be included in future work, alongside a validated, prospectively defined radiological response framework (whether an adapted RECIST/mRECIST approach or a formally validated HU threshold) and standardised CIRSE-based complication reporting from the outset. Longer follow-up is needed to assess the durability of response and recurrence risk. Further research should also evaluate the role of HU change as a quantitative imaging biomarker after cryoablation, ideally stratified by histological diagnosis. Defining tumour-specific

indications will help clarify where cryoablation provides the greatest clinical benefit.

Conclusion

Cryoablation appears to be a safe and effective minimally invasive treatment for selected musculoskeletal tumours. Most patients achieved a good, complete, or partial response, with a low complication rate. No major complications or procedure-related deaths were observed. Cryoablation may be useful for benign, cartilaginous, primary, and metastatic bone lesions, particularly when surgery is high risk or symptom control is the main aim. Hounsfield Unit reduction after treatment may provide a useful, though as yet non-validated, marker of radiological response. Further prospective studies with longer follow-up, validated response criteria, and standardised complication grading are needed to confirm these findings.

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Data availability Data is available to share on request.

Declarations

Ethical approval Local ethical committee approval was obtained. The study was conducted in accordance with the Declaration of Helsinki.

Consent to participate Not applicable.

Consent to publish Yes.

Declaration of AI use The authors declare the use of AI to improve the wording of this manuscript, with all outputs verified.

Conflicts of interest The authors declare that they have no conflict of interest.

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